

✓ Complete Exploratory Data Analysis

EDA - Exploratory Data Analysis Used to Explore, Summarize, Identify Anomalies, Test Assumptions, Detect Patterns, and check relationship between variables before applying statistical models or machine learning algorithms.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
path = "/content/Employee_dataset.csv"
```

```
data=pd.read_csv(path)
```

```
df=pd.DataFrame(data)
```

```
df.head()
```

	EmployeeID	recorddate_key	birthdate_key	orighiredate_key	termi
0	1318	12/31/2006 0:00	1/3/1954	8/28/1989	
1	1318	12/31/2007 0:00	1/3/1954	8/28/1989	
2	1318	12/31/2008 0:00	1/3/1954	8/28/1989	
3	1318	12/31/2009 0:00	1/3/1954	8/28/1989	
4	1318	12/31/2010 0:00	1/3/1954	8/28/1989	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.describe()
```

	EmployeeID	age	length_of_service	store_name	STATUS
count	49653.000000	49653.000000	49653.000000	49653.000000	49653.000000
mean	4859.495740	42.077035	10.434596	27.297605	2010.000000
std	1826.571142	12.427257	6.325286	13.514134	2.000000
min	1318.000000	19.000000	0.000000	1.000000	2006.000000
25%	3360.000000	31.000000	5.000000	16.000000	2008.000000
50%	5031.000000	42.000000	10.000000	28.000000	2011.000000
75%	6335.000000	53.000000	15.000000	42.000000	2013.000000
max	8336.000000	65.000000	26.000000	46.000000	2015.000000

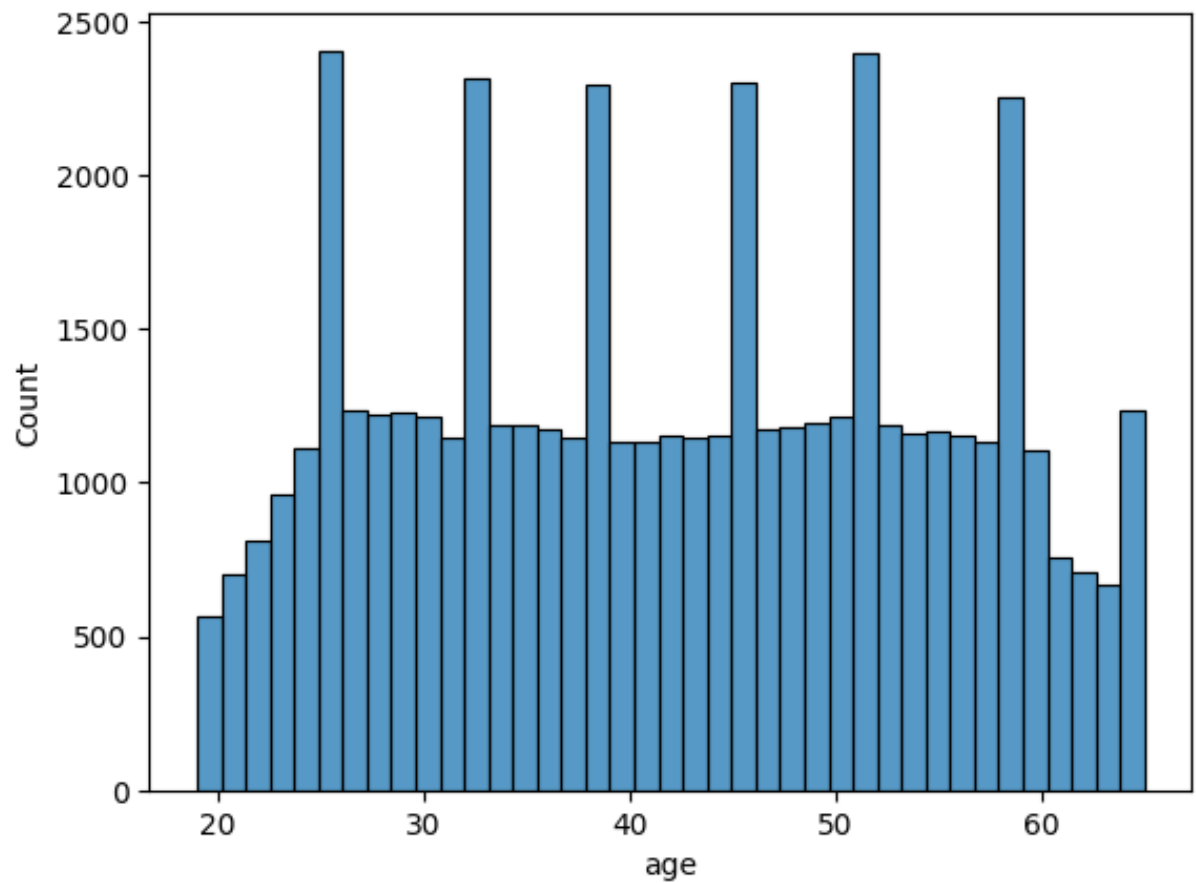
✓ UNIVARIATE ANALYSIS

Univariate Analysis is a type of Data Visualization where we can only visualize single variable at a time.

We Can Perform Univariate Analysis using Histogram on Numerical Variables.

```
sns.histplot(data['age'])
```

```
<Axes: xlabel='age', ylabel='Count'>
```

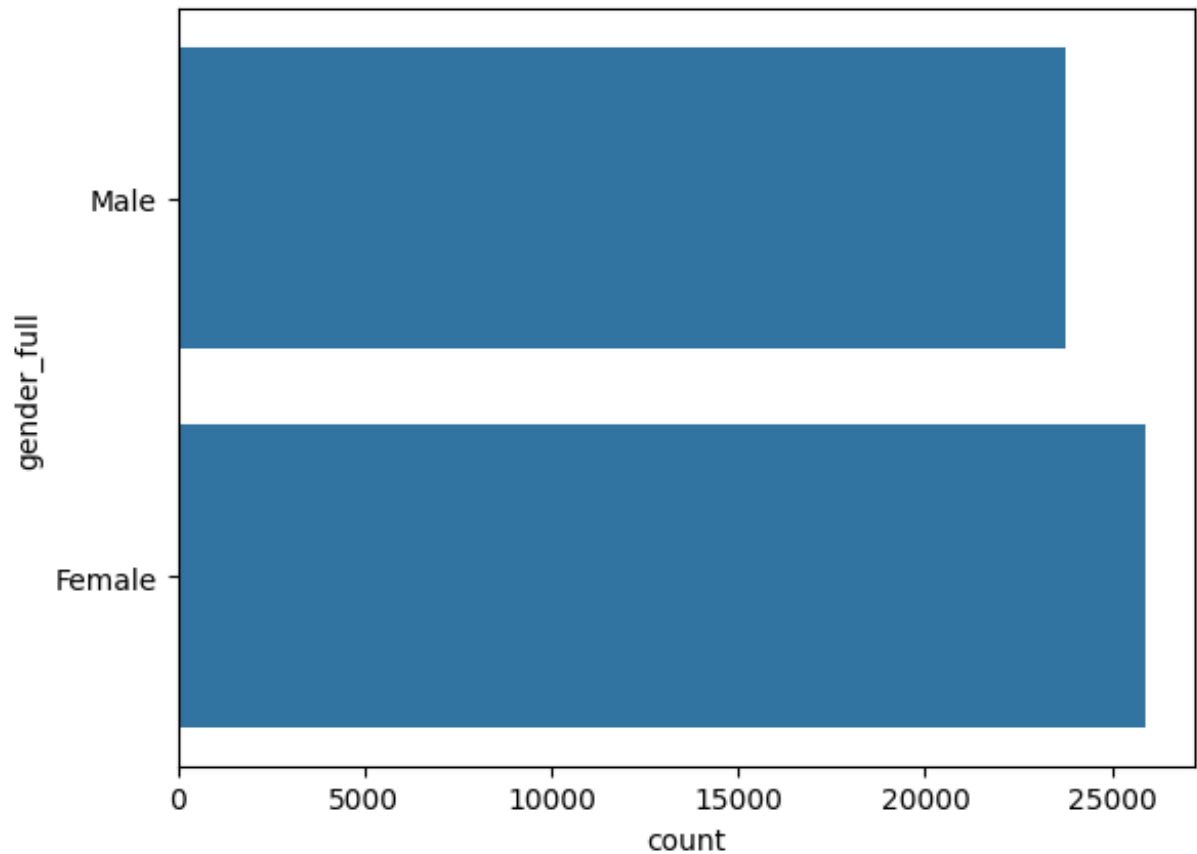


COUNTPLOT

For Categorical Data we can use countplot function using seaborn library

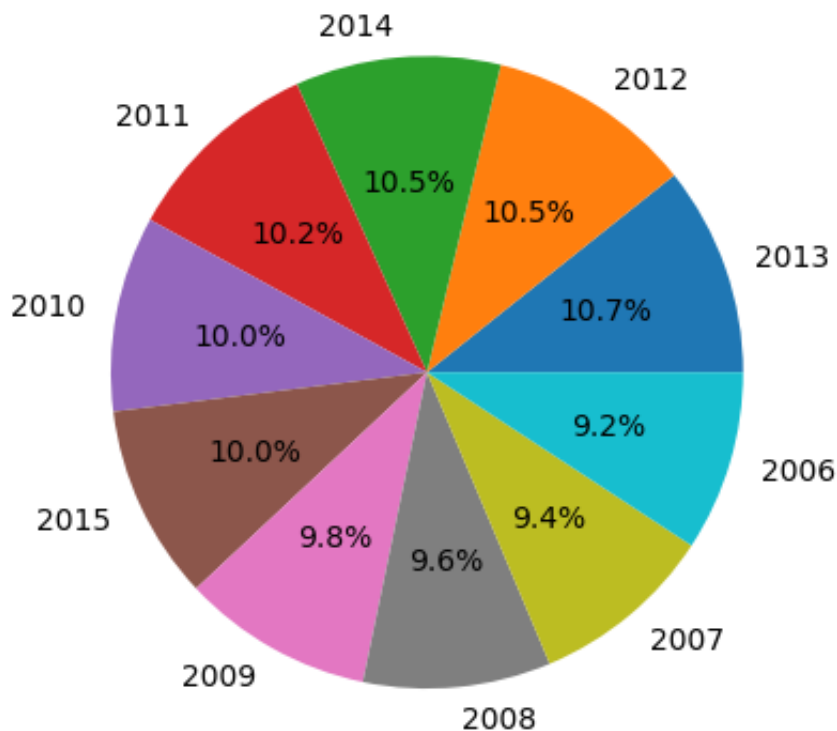
```
sns.countplot(data['gender_full'])
```

```
<Axes: xlabel='count', ylabel='gender_full'>
```



"Pie Chart" We can use Pie Chart to visualize the percentage of data belonging to each category using Plotly or Matplotlib

```
x = data['STATUS_YEAR'].value_counts()
plt.pie(x.values,
        labels=x.index,
        autopct='%1.1f%%')
plt.show()
```



✓ **BIVARIATE ANALYSIS**

BIVARIATE ANALYSIS used for simultaneous analysis of two variables.

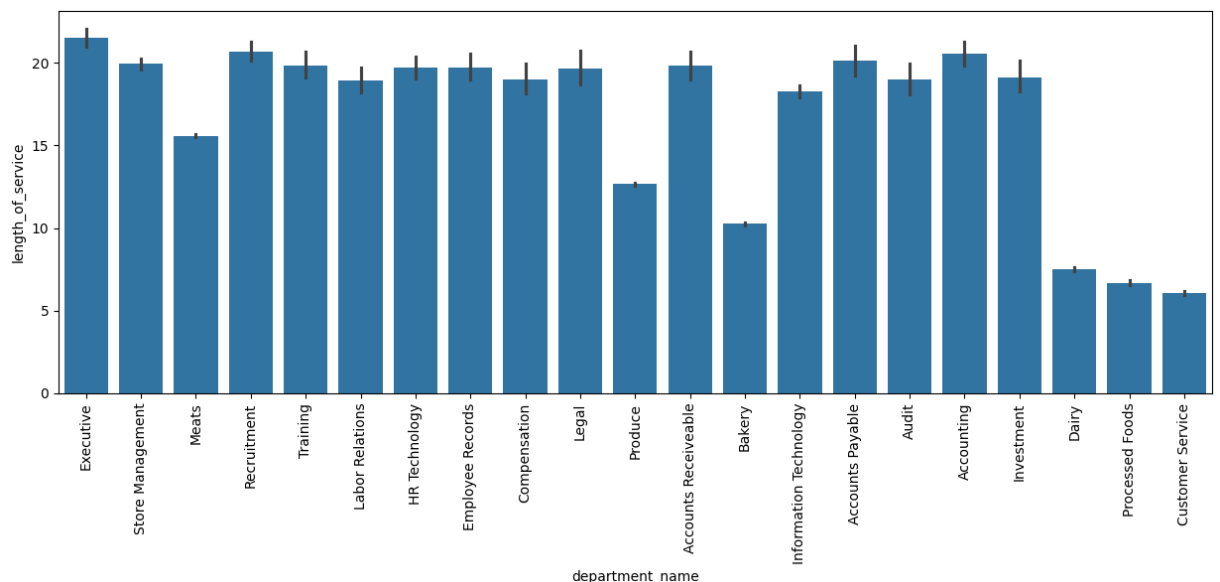
Major Types- 3

- Categorical vs Numerical
- Numerical vs Numerical
- Categorical vs Categorical

Categorical vs Numerical

```
"""Bivariate Analysis explores the concept of relationship whether
two variables impact each other."""
import matplotlib.pyplot as plt
plt.figure(figsize=(15, 5))
sns.barplot(x=data['department_name'], y=data['length_of_service'])
plt.xticks(rotation=90)
```

```
([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18,
19, 20],
[Text(0, 0, 'Executive'),
Text(1, 0, 'Store Management'),
Text(2, 0, 'Meats'),
Text(3, 0, 'Recruitment'),
Text(4, 0, 'Training'),
Text(5, 0, 'Labor Relations'),
Text(6, 0, 'HR Technology'),
Text(7, 0, 'Employee Records'),
Text(8, 0, 'Compensation'),
Text(9, 0, 'Legal'),
Text(10, 0, 'Produce'),
Text(11, 0, 'Accounts Receiveable'),
Text(12, 0, 'Bakery'),
Text(13, 0, 'Information Technology'),
Text(14, 0, 'Accounts Payable'),
Text(15, 0, 'Audit'),
Text(16, 0, 'Accounting'),
Text(17, 0, 'Investment'),
Text(18, 0, 'Dairy'),
Text(19, 0, 'Processed Foods'),
Text(20, 0, 'Customer Service')])
```

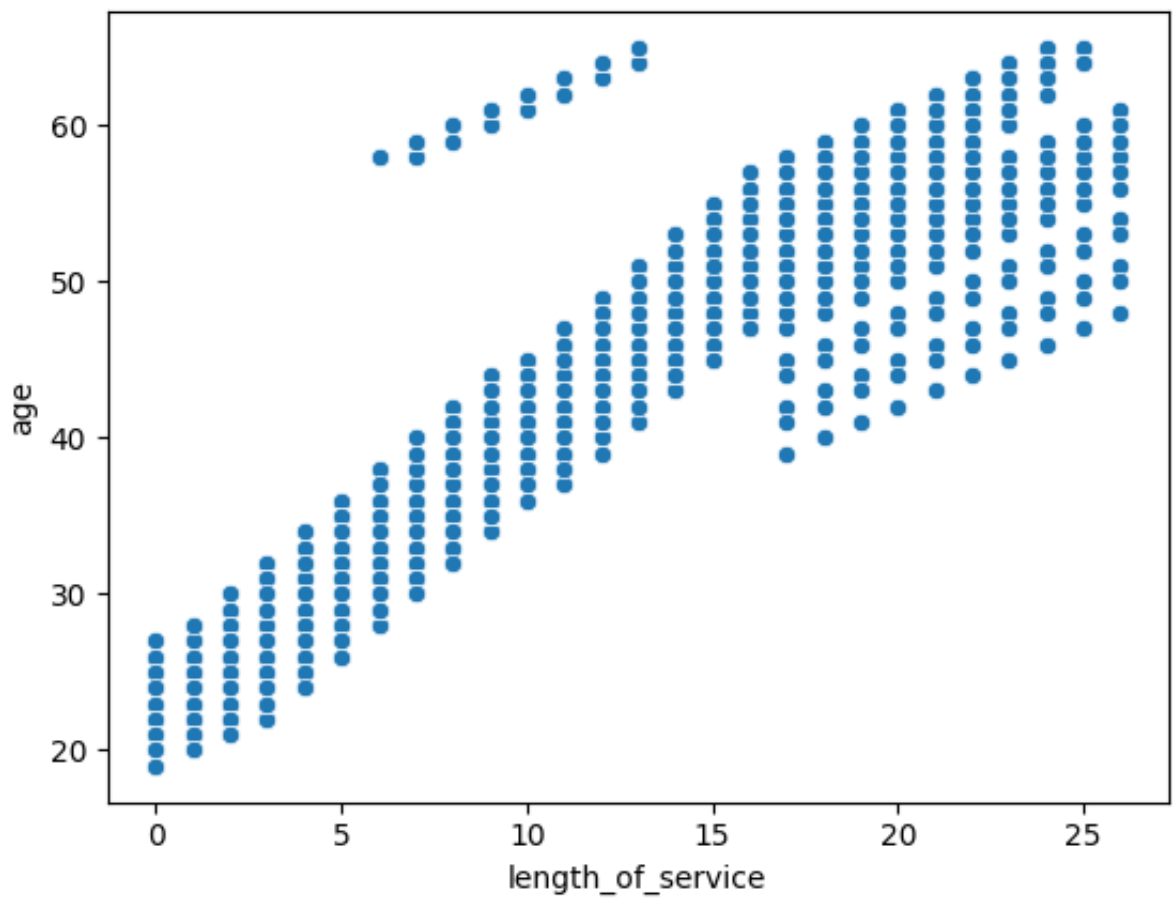


The Graph shows the length of service among different departments

Numerical vs Numerical

```
sns.scatterplot(x=data['length_of_service'],  
                y=data['age'])
```

<Axes: xlabel='length_of_service', ylabel='age'>

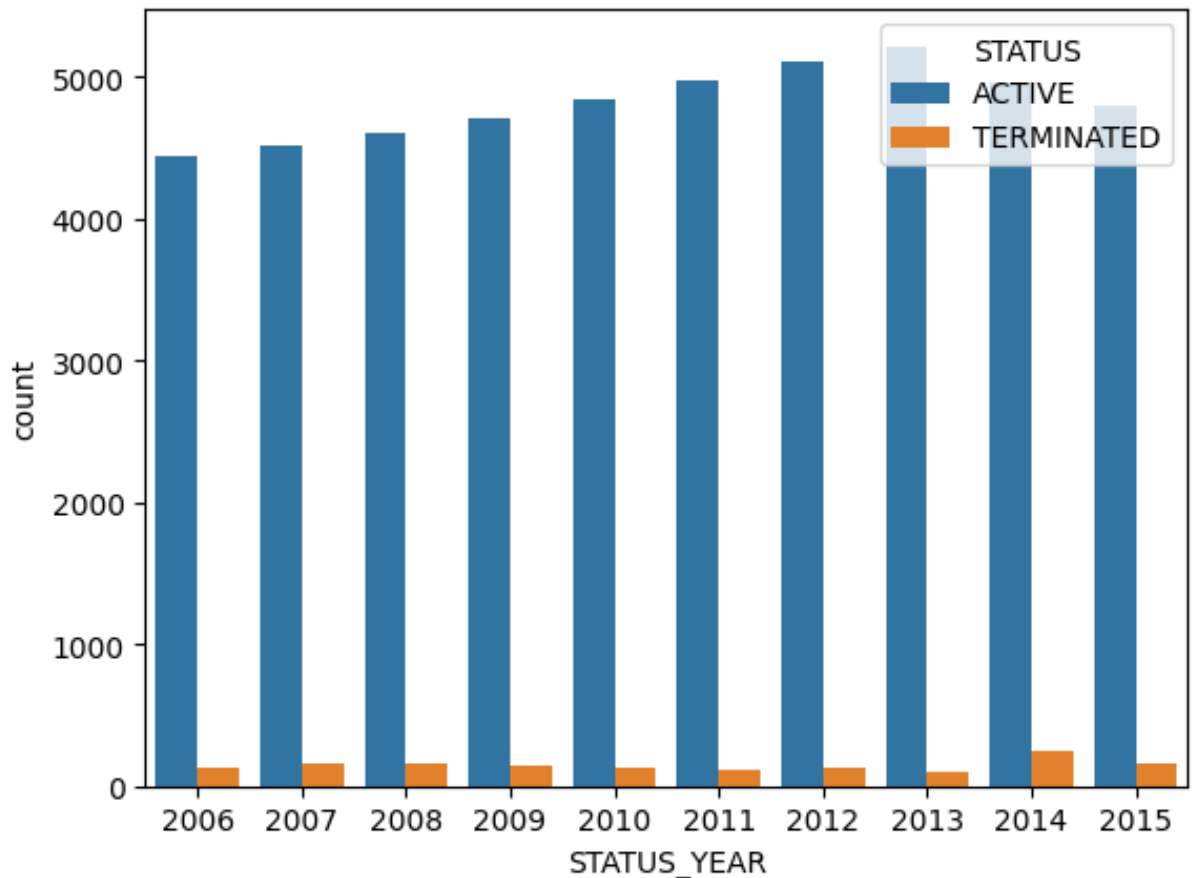


The Scatter Plot Displays the Length of Service of Employees in an office.

Categorical vs Categorical

```
sns.countplot(x='STATUS_YEAR', hue='STATUS', data=data)
```

```
<Axes: xlabel='STATUS_YEAR', ylabel='count'>
```



The graph shows the distribution of active and terminated employees for each year from 2006 to 2015.

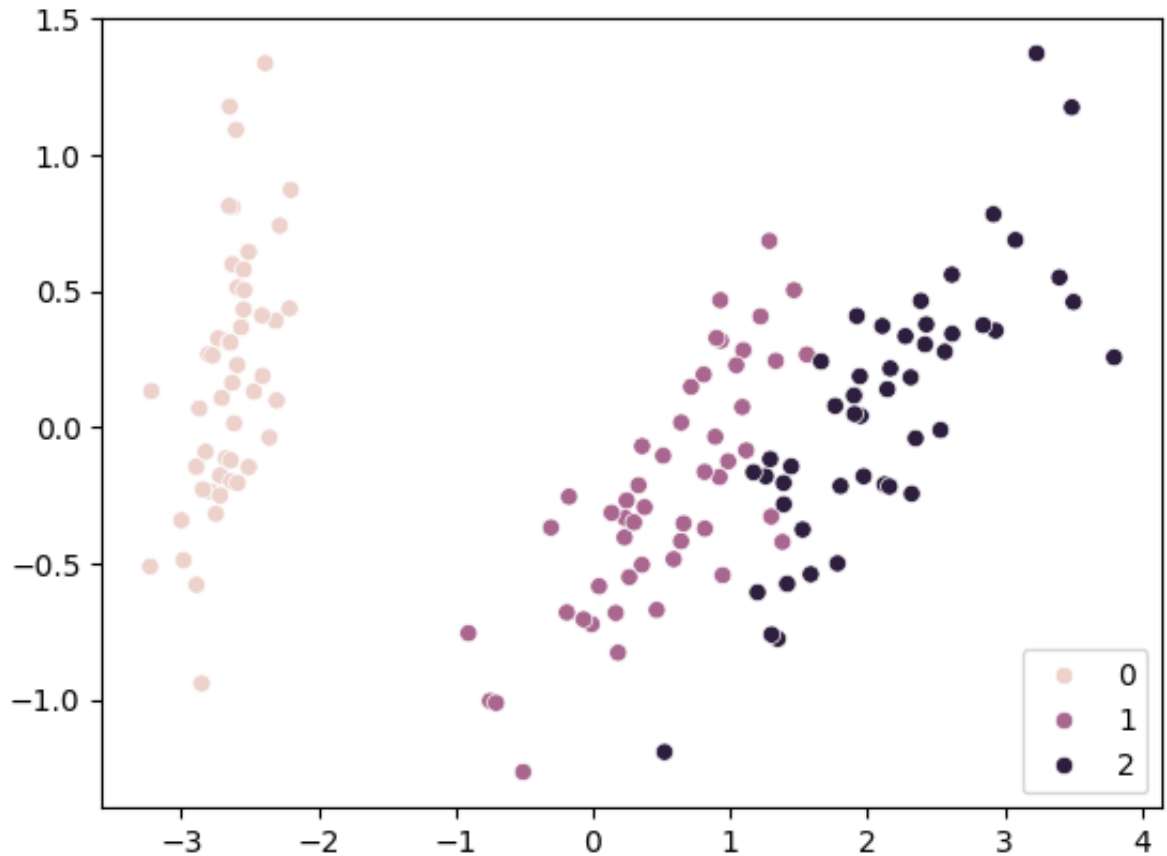
✓ ***Multivariate Analysis***

Multivariate Analysis is an extension of Bivariate Analysis and it involves multiple variables at the same time and examine patterns in multidimensional data.

PCA Principle Component Analysis


```
from sklearn import datasets, decomposition
iris = datasets.load_iris()
X = iris.data
y = iris.target
pca = decomposition.PCA(n_components=2)
X = pca.fit_transform(X)
sns.scatterplot(x=X[:, 0], y=X[:, 1], hue=y)
```

<Axes: >



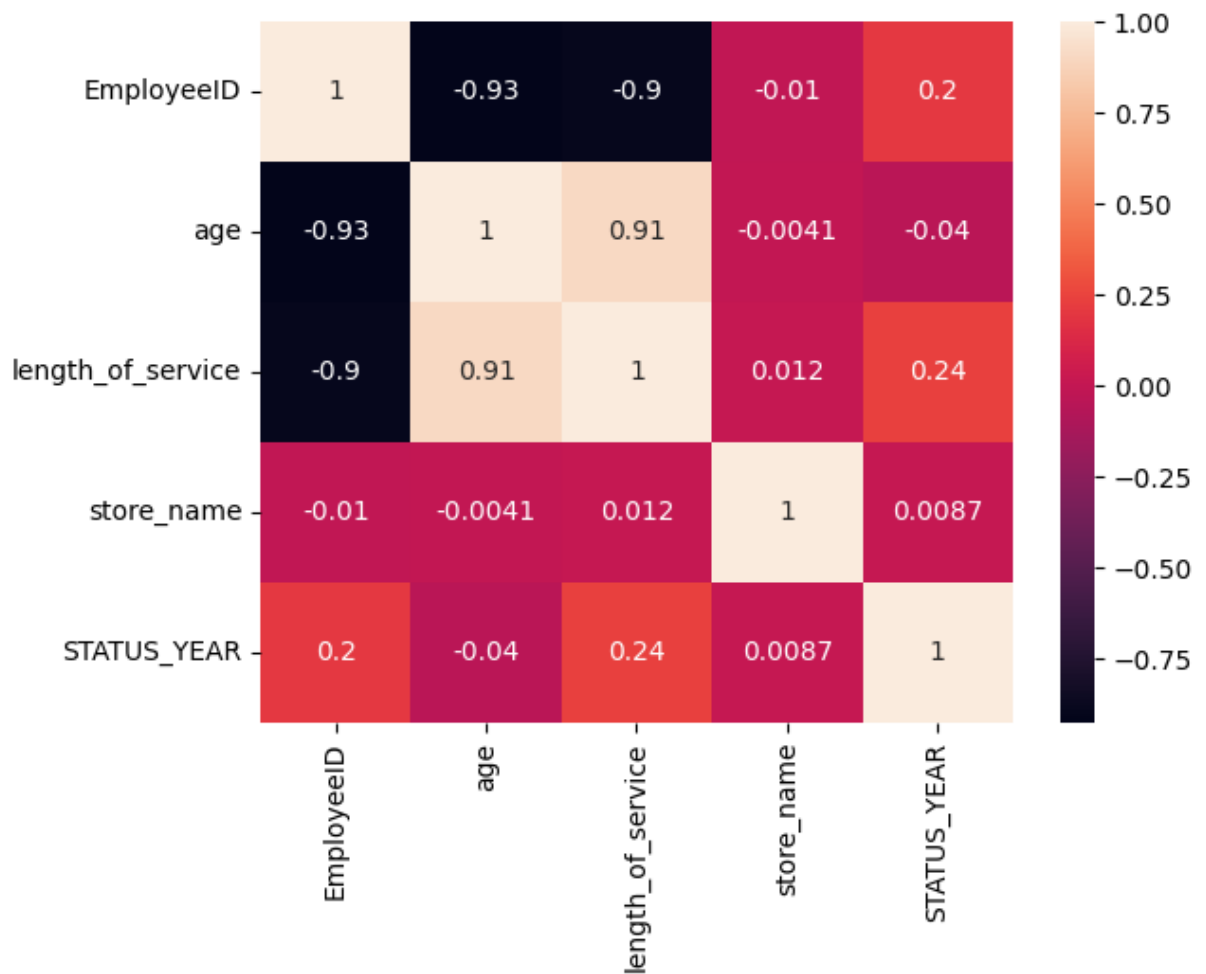
Principal Component Analysis (PCA) is mainly used for dimensionality reduction by projecting data onto directions that maximize the variance within the dataset.

Heatmap

Heatmap used to check the relationship and correlation

```
sns.heatmap(data.select_dtypes(include=['number']).corr(), annot=Tr
```

<Axes: >



The Values of Correlation can vary from -1 to +1 Where -1 refers to strong negativity and +1 refers to strong positivity.

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